

Preperatory R Course - Day One

author

Organization of this Course

Welcome! This course aims to teach the basics of R programming so you are prepared for the M.Sc. Forest and Ecosystem Sciences. Each day we are covering a different topic:

1. Getting started with R
2. Importing data, data frames, factors, data exploration
3. Using libraries, filtering data, rearranging data
4. Data visualization
5. Summary, Mini Project

At the end of the class you should be able to read a dataset into R, perform summary statistics, extract relevant information from the data, and create a simple visual representation.

Motivation

Many of you may be familiar with other means of statistical analysis (Excel, Stata...). Why do we use R instead? Programming may be intimidating at first, but R has many benefits:

- It is **reproducible**, meaning if someone has your code they can repeat every step of your analysis easily
- It is **free** instead of requiring a subscription or purchase
- It can deal with large datasets (hundreds of thousands of rows), including geospatial, genome, and other specialized data
- It is **open-source** with a large library of user-created packages that can extend its functionality. This makes R really flexible - for example, all teaching material for this class (including this pdf) is created in R.

R & RStudio

R is a **programming language** intended for statistical computing. We are using R in combination with **RStudio**, an integrated development environment (IDE). RStudio is a powerful tool that makes working with R much easier.

When opening RStudio, it by default has three panels (the **console**, the **environment**, and the **output** pane).

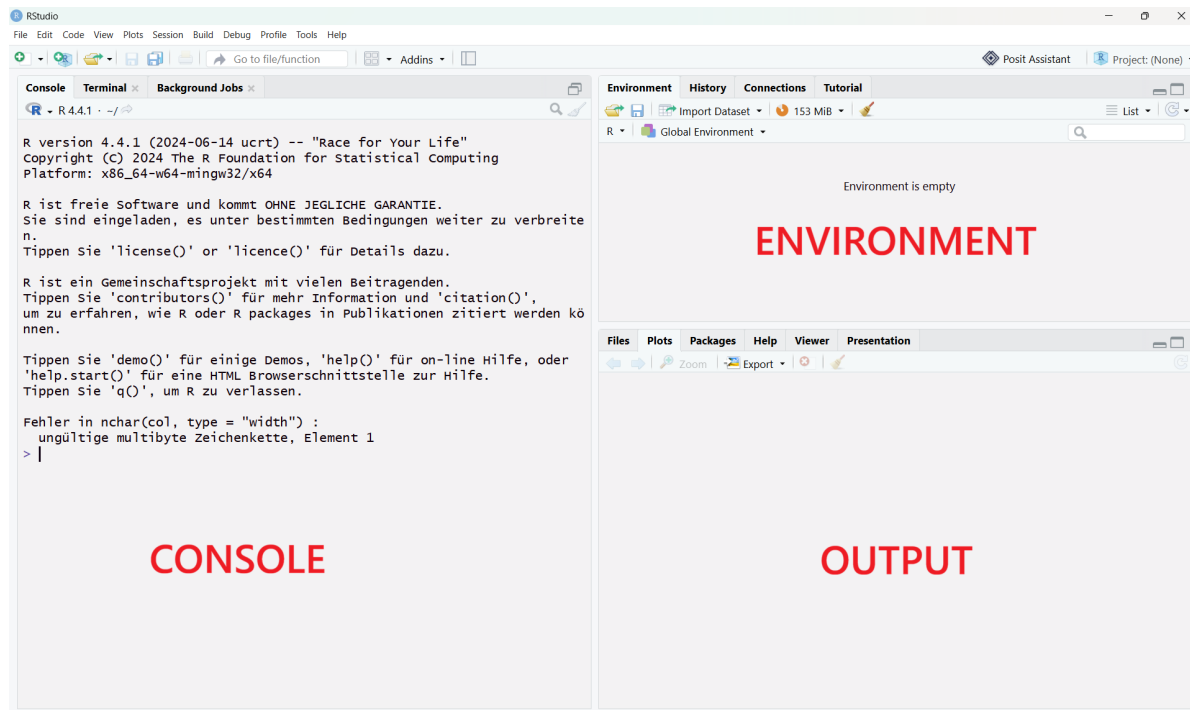


Figure 1: RStudio Panel Layout.

The environment shows any objects you have created (we will explain what this means later). The output window shows any plots you create, but also includes among other tabs a file explorer and a help window. The console is where your code is executed.

If you type the following code into the console and hit “enter”, you can see the result in the console:

```
1 + 1
```

```
[1] 2
```

If you close R, this code will be lost. It is therefore always advisable to save your results in an **R Code Script** (file ending .R). By clicking on *File -> New File -> R Script* in the top left corner, you open up a new panel, the **Script Panel**. Type any code in here and save it (*File -> Save*) and you can re-open it next time you open R!

 Note

Always try giving your files clear names. Avoid spaces and special characters other than `_` and `-` (including `Ä`, `Ö`...) in your file names, even if your system allows these file names.

Working with a Script

To run code from your R script, you place your cursor in the line you want to run. You then either click the **Run Button** in the top right corner of the script pane or click **CTRL + ENTER**.

You can see that the code is automatically pasted into the console. Any output for your code is also visible in the console.

 Tip

If you highlight part of a line and hit **CTRL + ENTER**, R will only run that part of the line. You can also highlight multiple lines the same way to run many lines at once.

Alongside your code you can write comments by starting your line with `#`.